

# Oussama Darouez

## Embedded Systems Engineer

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## Summary

Embedded Systems Engineer with hands-on experience designing firmware, real-time systems, and embedded networking protocols industrial and IoT projects. Skilled in bare-metal and RTOS-based development on various microcontrollers, driver design and CI/CD automation for production-grade firmware.

## Education

**National Institute of Applied Science and Technology (INSAT)**  
*Engineering Degree, Industrial Computing and Automation*

*Sept 2021 – Sept 2026*

## Experience

**Embedded Systems Engineering Intern**  
*Nicolaudie Group – LightingSoft AG*

*Meroux, France*

*March 2026 – Present*

- Developed a scalable **Modbus TCP/RS485 server** for STM32 and NXP LPC, supporting multiple concurrent clients.
- Integrated **Modbus triggers** into the show engine for real-time, dynamic scene parameter control.
- Built a robust **C++ Modbus client** with automated stress tests validating protocol reliability.
- Authored comprehensive **QTest** suites, strengthening regression coverage and system reliability.
- Migrated **Jenkins CI/CD** pipelines from Qt5 to Qt6, resolving automated test-build issues.

*Technologies:* C · C++ · Qt · QTest · STM32 · NXP LPC · Wireshark · Jenkins · SEGGER Ozone · FreeRTOS · Lizard Analysis

**Embedded Systems Engineer**  
*MareCustos*

*Tunis, Tunisia*

*June 2024 – Sept 2025*

- Built a **dual-slot OTA** firmware update system over Ethernet/USB with automatic fallback recovery.
- Prototyped reusable peripheral drivers for **Timers, Ethernet, Flash, ADC, and PCNT** modules.
- Designed custom **micro-ROS transport layers** for STM32, enabling reliable inter-device communication.
- Automated firmware build and testing workflows using **CI/CD** pipelines with GitHub Actions.
- Designed and debugged a custom **bootloader** with reset-cause handling for production robustness.

*Technologies:* C · C++ · STM32 · ESP32 · micro-ROS · Unity C · GitHub Actions

**Mechatronics Engineer**  
*Orange Tunisie*

*Tunis, Tunisia*

*May 2024 – July 2024*

- Optimized **FreeRTOS** task scheduling on ESP32 for reliable multitasking and efficient resource use.
- Restructured the **UART/Ymodem** interface for reliable command reception and task triggering.
- Established a robust **test bench** to streamline robot functionality validation.

*Technologies:* ESP32 · FreeRTOS · UART · Ymodem · Bash

**Robotics Instructor**  
*Discovery Club Junior*

*Manouba, Tunisia*

*May 2023 – May 2025*

- Delivered hands-on **PCB design** training with KiCad, from schematic to prototype.
- Guided students through **digital logic design** and ATmega328P register-level programming.
- Assisted students with mechanical design and robot mechanics brainstorming using SolidWorks.

*Technologies:* KiCad · SolidWorks · ATmega328P · Robotics

## Projects

**OTA-Enabled Smart Home Devices**

*08/2025 – 09/2025*

- Prototyped IoT devices with **HTTPS OTA** updates and SmartConfig Wi-Fi provisioning.
- Implemented **non-volatile storage** management for reliable device parameter retrieval.
- Automated firmware build, flash, and deployment using **Bash** and **GitHub Actions** for ESP-IDF.

*Technologies:* ESP-IDF · Wi-Fi · GitHub Actions · Bash

**Bare-Metal Drivers**

*01/2024 – 05/2024*

- Developed low-level **bare-metal drivers** for STM32, PIC and Atmega: GPIO, ADC, Timer, and EXTI peripherals.
- Implemented full drivers for **LIS3DSH (SPI)** and **MMA7660 (I2C)** accelerometers.
- Built a configurable **brushless motor control** system using PWM and ADC feedback.

*Technologies:* STM32F4 · PIC16F · Atmega328P · SPI · I2C

**Autonomous Robots, Eurobot**

*09/2022 – 05/2023*

- Engineered servo drivers and **GPIO HATs** with KiCad, validated through real-world testing.
- Implemented **PID motor control** and odometry on STM32 for precise robot movement.
- Set up **ROS1** communication between Raspberry Pi and STM32 across multi-board systems.

*Technologies:* STM32 · ROS1 · KiCad · 3D Printing

## Skills

**Microcontrollers:** STM32, ESP32, PIC, ATmega328P, Raspberry Pi, NXP LPC.

**Protocols:** Ethernet, TCP/IP, UART, SPI, I2C, USB-CDC, CAN, Modbus, RS485, Wi-Fi.

**RTOS & Tools:** FreeRTOS, Qt, SEGGER Ozone, Wireshark, KiCad, SolidWorks.

**Languages & CI/CD:** C, C++, Python, Java, Bash, Git, GitHub Actions, Jenkins, Docker.

## Languages

**English:** Proficient, **French:** Proficient, **Arabic:** Native.